

Research Paper:

Small Object Detection Algorithm Based on Improved Attention Mechanism and Feature Fusion of YOLOv8

Mingxing Fang^{*1,*2,*3,†}, Xinyu Rui^{*1}, Hongyu Cheng^{*1}, Xinke Liu^{*1},
Jinhua She^{*4}, Youwu Du^{*5}, and Haoran Tan^{*6}

^{*1}School of Physics and Electronic Information, Anhui Normal University
No.189 Jiuhua South Road, Yijiang District, Wuhu, Anhui 241002, China
E-mail: mxfang@mail.ahnu.edu.cn

^{*2}Anhui Engineering Research Center on Information Fusion and Control of Intelligent Robot
No.189 Jiuhua South Road, Yijiang District, Wuhu, Anhui 241002, China

^{*3}Anhui Provincial Joint Key Laboratory on Information Fusion of Intelligent Automotive Cabin
No.189 Jiuhua South Road, Yijiang District, Wuhu, Anhui 241002, China

^{*4}School of Engineering, Tokyo University of Technology
1404-1 Katakuramachi, Hachioji, Tokyo 192-0982, Japan

^{*5}School of Electrical and Information Engineering, Jiangsu University of Technology
No.1801 Zhongwu Road, Changzhou, Jiangsu 213001, China

^{*6}National Engineering Research Center of Robot Visual Perception and Control Technology, Hunan University
No.2 Lushan South Road, Yuelu District, Changsha, Hunan 410082, China

[†]Corresponding author

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Addressing the challenges in small object detection, particularly the issue that small objects often suffer from a lack of sufficient semantic information and are highly susceptible to background noise, this paper proposes an innovative algorithm, namely YOLOv8-FE. Firstly, to enhance the network's sensitivity to small object detection, a P2-scale detection layer specifically designed for small objects is integrated into the model. Secondly, addressing the potential information loss during downsampling in traditional convolutional layers, an innovative downsampling module named RFAC-SPD is designed, aiming to more effectively capture and utilize features of small objects, thereby assisting the model in improving performance. Additionally, to mitigate the interference from background noise and strengthen the network's ability to focus on object information, the study builds the C2f-CBAM module based on the convolutional block attention module (CBAM). Moreover, to fully integrate low-level feature information, minimize the loss of underlying detail information, and further enhance the network's representational capability, an enhanced path aggregation network is proposed, significantly improving the effectiveness of network feature fusion. Experiments on the dataset VisDrone2019 show that the YOLOv8-FE algorithm exhibits superior performance and detection efficiency. Compared to the baseline algorithm YOLOv8n, its mAP50 and mAP50-95 have increased by 8.3% and 5.3%, respectively. Furthermore, with an inference speed of 77 frames per second, YOLOv8-FE meets real-time requirements, thereby validating the advancement and effectiveness of the proposed improvement algorithm. Furthermore, generalization experiments con-

ducted on the DOTA and Caltech Pedestrian datasets demonstrate that the improved model achieves an increase of 2.7% and 6.8% in mAP50, respectively, fully validating the generality of the proposed model.

Keywords: small object detection, YOLOv8, RFAC-SPD, EPAN

1. Introduction

In recent years, the rapid advancement of computer vision and artificial intelligence has led to increasingly significant impacts of deep learning-based small object detection techniques in practical applications, including security monitoring systems, intelligent transportation systems, and unmanned aerial vehicle search and rescue missions.

Hand-designed features such as Haar features [1] and histogram of oriented gradients [2] are used by traditional object detection algorithms to characterize the target region. However, for small objects with limited feature information, these hand-designed features sometimes fail to adequately describe the target, resulting in lower detection accuracy. In addition, traditional object detection algorithms suffer from high computational complexity, which makes it difficult to meet the criteria of scenes with high real-time requirements. In contrast, deep learning-based object detection methods can capture the complex and subtle features of small objects by automatically learning the representation of features, thereby improving the detection accuracy of such objects. In addition, deep learning methods usually have high computational efficiency and real-



time performance, which can realize faster detection speed and meet the needs of practical applications.

The most popular target detection methods in current deep learning are mainly categorized into two main groups: the two-stage method and the one-stage method. Specifically, the two-stage method involves an initial stage of generating region proposals, followed by a classification process utilizing a convolutional neural network. The notable characteristic of this method is its high accuracy, albeit at a relatively slower speed. Common two-stage object detection methods include R-CNN (regions with convolutional neural networks) [3], Fast R-CNN [4], Faster R-CNN [5], and Mask R-CNN [6]. Conversely, the One-stage method operates without the need for generating region proposals, instead directly extracting features from the network to predict both object classifications and locations. Consequently, it demonstrates a superior detection speed. The YOLO (You Only Look Once) series and SSD (single shot multibox detector) [7] are examples of popular One-stage object detection algorithms.

These object detection algorithms are typically designed for natural scenes and show excellent detection results on datasets such as MS COCO [8], which consists mainly of images containing larger objects. However, for most images containing small objects, applying these algorithms directly to small object detection can lead to degraded performance of the detector due to the low pixel count, simplistic features, complex backgrounds, and numerous interfering factors exhibited by small objects. This results in challenges such as decreased accuracy and elevated miss detection rates.

The algorithm is adjusted to accommodate the unique attributes of small objects; this paper proposes a YOLOv8-FE algorithm based on the YOLOv8 network that can elevate the precision of detecting small objects. The key contributions of this algorithm are outlined below:

- Add a P2-scale detection layer to increase attention to small objects and enhance the network's capability of recognizing small targets.
- Introduce the RFAC-SPD module to enable the network to fully learn target features while preserving more target information during downsampling.
- Design the C2f-CBAM module to strengthen the network's concentration on important features, thus enhancing the reliability of object detection.
- Propose the enhanced path aggregation network (EPAN) to optimize cross-scale connections and upgrade the precision of object detection.

2. Related Works

2.1. Object Detection

At present, object detection algorithms are mainly classified into two-stage object detection algorithms based on region proposals and one-stage object detection algorithms

based on regression. Among them, the two-stage object detection algorithm R-CNN is a pivotal model that first applied deep learning to object detection, but its redundant computation of overlapping bounding box features can lead to slower detection speeds [3]. Subsequently, He et al. [9] proposed SPPNet (spatial pyramid pooling networks), in which the spatial pyramid pooling (SPP) layer can convert inputs of different sizes into fixed-size outputs to avoid repeated computations. The RoI (region of interest) Pooling layer adopted by Fast R-CNN [4], and the region proposal network introduced by Faster R-CNN [5] further improved detection accuracy and increased detection speed. The proposal of feature pyramid networks (FPNs) addresses the shortcomings of the Faster R-CNN object detection algorithm in handling multi-scale variations [10]. It employs multi-scale feature fusion to enhance the representation of shallow features and significantly improves the detection capability for small objects by extracting large-resolution shallow features.

One-stage object detection algorithms offer faster detection speeds compared to two-stage detection algorithms. The YOLO algorithm, proposed by Redmon et al., treated object detection as a regression problem for the first time, enabling the prediction of object locations and categories in an image through a single forward pass [11]. This approach simplified the detection process and increased speed. YOLOv2 employed Darknet-19 as the feature extraction network and applied techniques such as batch normalization, anchor boxes, and multi-scale training, which improved the model's convergence speed, stability, and detection capabilities [12]. YOLOv3 adopted a deeper and more powerful Darknet-53 as the feature extraction network and introduced residual connections, enhancing the network's feature extraction ability [13]. YOLOv4 integrated the concept of cross stage partial network into Darknet-53, upgrading it to CSPDarknet-53 as the backbone network of YOLOv4, which strengthened feature representation capabilities. Additionally, YOLOv4 introduced the SPP and the path aggregation network structure, further boosting the algorithm's performance [14]. Jocher et al. proposed YOLOv5, which underwent lightweight processing on the network and provided five different size versions for users to choose from, adapting to various application scenarios. YOLOv6 [15] introduces a novel backbone network, the EfficientRep Backbone, alongside the Rep-PAN Neck feature pyramid structure, significantly enhancing inference speed while maintaining high performance. YOLOv7 [16] proposes efficient network architectures such as extended efficient layer aggregation network (ELAN) and enhanced ELAN, which significantly improve detection performance. YOLOv8 focuses on striking an optimal balance between accuracy and speed, adopting advanced backbone and neck architectures, as well as detection heads, successfully achieving a notable increase in accuracy while maintaining high speed. As computer vision technology continues to evolve, the YOLO series of models will undergo continuous improvement and optimization.

2.2. Small Object Detection

The widespread application of deep learning technologies has propelled recent advancements in the field of general object detection. However, object detection algorithms designed for natural scenes often exhibit less than satisfactory performance when tasked with detecting small objects. In light of this, numerous researchers have conducted extensive and in-depth studies to address the challenges faced in small object detection. Wang et al. [17] designed a multiple-in-single-out attention encoder and input the outputs of the last two stages of the backbone network into the encoder for feature fusion, thereby enhancing the local and global representations of the feature maps and optimizing the detection results for smaller objects. Jing et al. [18] introduced a feature fusion module, a visual perception module, and a feature encoding module, enabling the network to retain more fine-grained features. This enhancement ensures improved attention to small objects even at low resolutions, while simultaneously reducing network parameters and computational overhead. Zhang et al. [19] designed a dual-backbone Swin-Transformer network utilizing the dense higher-level composition model, and integrated it with FPN to establish a dedicated small object feature extraction network. This network is adept at capturing a more robust feature representation for small objects, while preserving the semantic information of larger objects. Ji et al. [20] proposed an attention module and an expanded field-of-perception block, which integrates the features of small objects with contextual information, thereby enhancing the robustness of feature extraction and strengthening the network's capability in locating and identifying small objects. Zhu et al. [21] incorporated transformer prediction heads and the convolutional block attention module (CBAM) into the YOLOv5 framework to capture both global and intricate contextual information. Li et al. [22] proposed an efficient and robust convolution residual squeeze-and-excitation module, which facilitates seamless transmission of information features and augments the ability of feature extraction. Wang et al. [23] introduced the effective receptive field module, which can enhance the receptive field of convolutional kernels while preserving local details, allowing the model to leverage more contextual information. Meng et al. [24] revised the detection head and anchor box parameters of the network to capture more comprehensive features of small objects, thereby enhancing its capability to perceive dense clusters of small objects. Additionally, they introduced an attention-based fusion module, termed ECF, which is designed to extract both global and local contextual information, ultimately improving the accuracy of small object detection in complex scenarios. Yang et al. [25] re-designed the classification and regression loss functions of the model based on YOLOX to better enhance the network's classification and localization capabilities.

In summary, researchers generally tend to adopt a series of strategies to enhance the feature representation capability of small targets, which include strengthening feature representation, fusing multi-scale feature information,

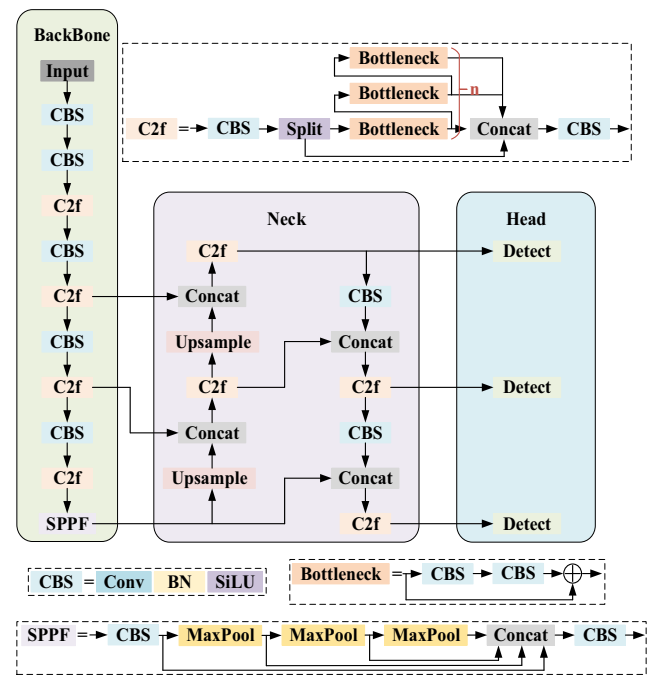


Fig. 1. YOLOv8n network architecture.

and optimizing loss functions, among others. These efforts aim to achieve effective integration with mainstream object detectors, thereby improving the detection capability of algorithms for small targets. Despite extensive research conducted by numerous researchers in the field of small object detection, accurately detecting small objects remains a key focus and a challenging area that requires further investigation.

2.3. YOLOv8

YOLOv8 is a further optimized object detection model within the YOLO series, boasting improved speed and accuracy compared to previous versions. Compared to earlier YOLO versions, YOLOv8 operates faster and achieves higher accuracy. The algorithm is classified into five versions based on its scale size (n, s, m, l, and x). For the comprehensive consideration of real-time, accuracy and model size of target detection, YOLOv8n, which has fewer parameters and faster speed, is used as the base model for improvement. Its network architecture diagram is shown in Fig. 1.

The YOLOv8n model architecture can be broadly categorized into four components: Input, Backbone, Neck, and Head. In the Input stage, adaptive image resizing is utilized to adjust the input dimensions, while mosaic data augmentation is employed to bolster the model's robustness. The Backbone employs the CSPDarkNet structure for the extraction of image features, and the C2f module is incorporated to enhance the model's gradient flow by introducing additional skip connections and split operations. The Neck section further refines features and facilitates more comprehensive feature fusion through cross-layer connections from top to bottom and vice versa, leveraging the FPN and

PAN structures. Finally, the Head section adopts a decoupled head structure, converting feature maps into the necessary category, location, and confidence information for object detection, effectively boosting model performance.

While YOLOv8 demonstrates impressive performance on natural scene image datasets, it is not specifically designed for small object detection tasks, leaving significant room for improvement in this area. YOLOv8-NDTiny [26] and BDAD-YOLO [27] replaced the convolutional modules in the C2f module with deformable convolutions, enabling the model to adapt to complex scenes. Wang et al. [28] introduced a small object detection layer into YOLOv8 and integrated the efficient multi-scale attention module with the C2f module, designing the C2f-E structure to enhance the network's focus on small objects in complex backgrounds. Li et al. [29] introduced a mixed local channel attention mechanism into the C2f structure, allowing the model to pay more attention to regions containing small objects. Zhao and Chen [30] used the ASF-DySample structure to enhance the fusion of high-dimensional information from deep feature maps and detailed information from shallow feature maps. Ning et al. [31] to better preserve the detailed information of low-resolution images and small objects, replaced the original strided convolutions in the backbone network with SPD-Conv. Yi et al. [32] designed the attention-guided bidirectional FPN, adding a top-down path to convey high-level semantic feature information and connecting the first and last nodes of the intermediate feature layers for better fusion. Wu et al. [33] designed the high-resolution FPN, which enhances small object detection by effectively integrating features of different scales and leveraging the unique properties inherent in each scale. Zhang [34] have improved the neck of YOLOv8 by tailoring a detection head specifically for small-sized targets and designing the sandwich-fusion module with a large receptive field to effectively capture spatial features from the lower layers of the backbone network.

The aforementioned research has once again emphasized the crucial role of feature extraction and feature fusion in small target detection. In light of this, this paper proposes the YOLOv8-FE algorithm, which improves the detection performance for small targets by improving the extraction and fusion processes of their features. Experimental results indicate that the proposed algorithm significantly enhances the detection effectiveness for small targets.

3. Research Methods

According to the characteristics of small targets, this paper proposes a YOLOv8-FE small target detection algorithm. Building upon the YOLOv8 model, an augmented P2-scale detection layer is integrated, significantly enhancing its ability to capture detailed information about small objects. Additionally, the convolutional downsampling process in the shallow layer of the backbone network is substituted with the RFAC-SPD module, thereby enabling

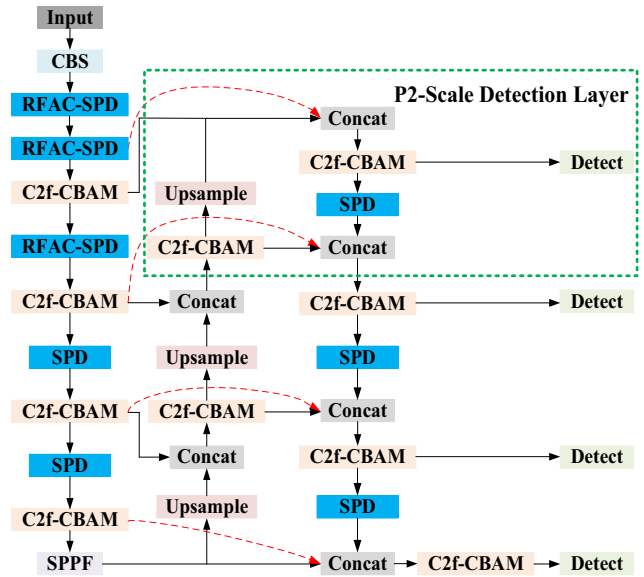


Fig. 2. YOLOv8-FE network architecture.

a more efficient extraction and processing of image features. To enhance the representation of significant features and positions, the CBAM module is integrated into the C2f module, forming the C2f-CBAM module. Subsequently, the original PAN structure is replaced by the EPAN to achieve comprehensive integration of multi-scale features. The architecture of the YOLOv8-FE is illustrated in Fig. 2.

3.1. Adding a P2-Scale Detection Layer

The YOLOv8n raw model employs an input resolution of 640×640 pixels and generates feature maps of sizes 80×80 , 40×40 , and 20×20 through its three feature layers (P3, P4, and P5) in its multi-scale detection architecture. Among these layers, the P3 layer is responsible for detecting small targets and has the highest spatial resolution, equivalent to the resolution of the original image after eight-fold downsampling. However, since small targets typically occupy only a small number of pixels in the original image, this can easily lead to the loss of feature information for small targets during the deep feature extraction process. Therefore, despite the high resolution of the P3 layer, the original YOLOv8n model still encounters significant challenges in capturing the feature information of small targets, thereby affecting its ability to detect them.

To enhance the detection capability for small objects, the YOLOv8n model is augmented with a P2-scale detection layer, specifically tailored for small objects and featuring an output image resolution of 160×160 , thereby improving the detection performance for such objects. As depicted in Fig. 2, a feature fusion layer and a detection head are incorporated into the original model to constitute the P2-scale detection layer. This enhancement augments the semantic information and feature representation capabilities for small objects, allowing the model to detect and locate them with greater precision, ultimately improving the accuracy of small object detection.

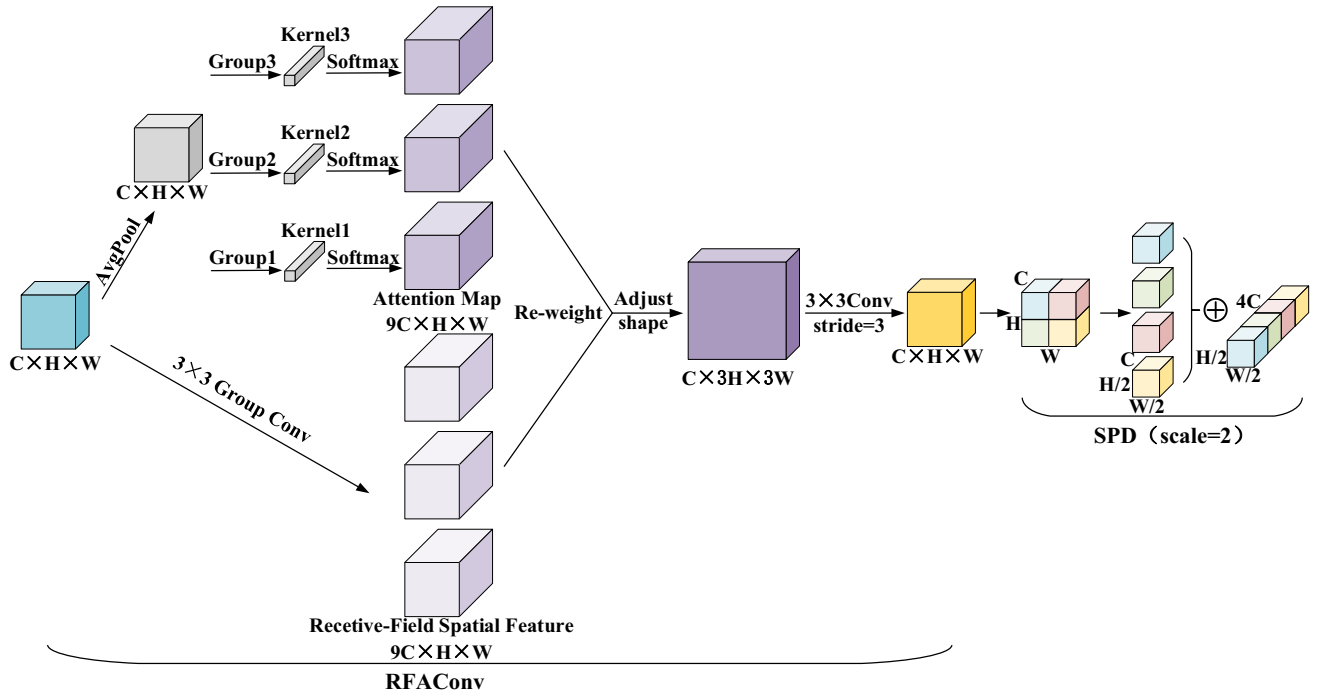


Fig. 3. Structure of RFAC-SPD module.

3.2. Proposing the RFAC-SPD Module

During the downsampling process of convolutional layers in YOLOv8, as the resolution decreases, some detailed information of small objects may be lost, rendering their identification and localization challenging in subsequent processing stages. Moreover, for small objects, their feature information is inherently scarce, and the decrease in resolution further exacerbates the issue of insufficient information. Given this, to avoid the decline in the network's ability to detect small objects due to the loss of fine-grained information, this paper proposes the use of the space-to-depth (SPD) [35] layer as a replacement for the downsampling convolution in the original network. Furthermore, in order to ensure that the feature maps obtained after downsampling contain more critical features related to small objects, this paper introduces the receptive-field attention convolutional (RFACConv) [36] layer and designs a novel feature downsampling strategy, namely the RFAC-SPD downsampling module, as illustrated in **Fig. 3**. This module is applied to the shallow layers of the backbone network. By enhancing the feature extraction capability of these shallow layers, the network is able to capture more crucial details of small objects and effectively transmit these rich feature representations to subsequent network layers, providing useful information for subsequent tasks.

Within the realm of object detection, RFACConv emerges as an innovative convolutional operation method that integrates spatial attention with receptive field characteristics to efficiently tackle intricate patterns and fine details in images. Unlike traditional convolutional operations, which primarily concentrate on spatial dimensions, RFACConv devotes special attention to spatial features within the re-

ceptive field. This enables the network to comprehend and process localized regions of the image more effectively, thereby enhancing the precision of feature extraction. Furthermore, by incorporating an attention mechanism, RFACConv possesses the capability to adjust the parameters of convolutional kernels with flexibility, offering tailored processing for diverse regions and thereby augmenting the model's adaptability. The computation of RFACConv is depicted in Eq. (1).

$$F = \text{Softmax} \left(g^{1 \times 1}(\text{AvgPool}(X)) \right) \times \text{ReLU} \left(\text{Norm} \left(g^{k \times k}(X) \right) \right) \quad (1)$$

In Eq. (1), X denotes the input feature map, F denotes the output feature map, $g^{k \times k}$ denotes the grouped convolution with a size of $k \times k$, Norm indicates normalization, Softmax and ReLU represent activation functions, and $\text{AvgPool}(\cdot)$ represents average pooling.

SPD is a space-to-depth layer that converts spatial dimensions into channel dimensions, thereby augmenting the channel count at the expense of spatial resolution. This approach better preserves spatial information, especially for small objects and detailed parts, thereby enhancing feature representation capabilities and improving the model's perception of details.

The RFAC-SPD module sequentially processes the feature maps through RFACConv and SPD, thereby allowing the network to leverage both the receptive field attention mechanism and the space-to-depth transformation technique. This facilitates the model in capturing and utilizing detailed feature information of small objects more effectively, achieving efficient downsampling, and ultimately enhancing the accuracy of small object detection.

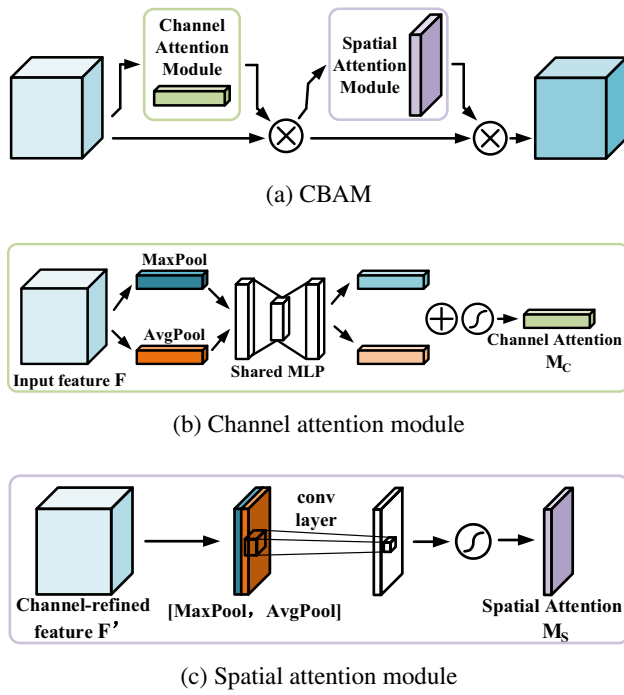


Fig. 4. Structure of CBAM. (a) The structural diagram of CBAM. (b) The channel attention module in CBAM. (c) The spatial attention module in CBAM.

3.3. Improvement of the C2f Module

In real-world scenarios, small objects frequently face challenges in being distinctly separated from intricate backgrounds due to their minute size and low resolution. In response, an attention mechanism is incorporated to mitigate background noise and interference while enhancing the emphasis on the feature information pertaining to small objects within images.

CBAM [37] is a concise yet potent attention module for feed-forward convolutional neural networks, as shown in **Fig. 4**, which has two sequential sub-modules: channel and spatial. The channel sub-module focuses on capturing the interdependencies among channels, whereas the spatial sub-module is dedicated to modeling the spatial correlations between pixels. During the learning process, it automatically assesses the importance of each feature channel and feature space, ultimately amplifying the critical features while diminishing those of lesser significance for the given task. The relevant computational expressions are outlined below:

$$F'' = M_S (M_C(F) \otimes F) \otimes (M_C(F) \otimes F). \quad (2)$$

In Eq. (2), F represents the input feature map, $\otimes$ denotes element-wise multiplication, and $M_C(\bullet)$ represents the feature map that emerges after the input has traversed the channel attention module, with its precise formulation detailed in Eq. (3). Similarly, $M_S(\bullet)$ denotes the feature map that results from the input passing through the spatial attention module, as specified in Eq. (4). F'' represents the new feature obtained after passing through CBAM.

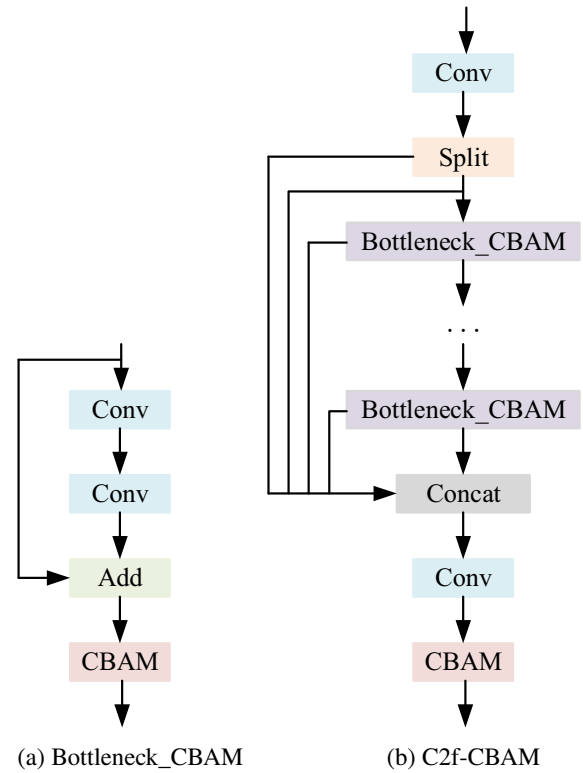


Fig. 5. Structure diagram of the C2f-CBAM module.

$$M_C(F) = \sigma(MLP(AvgPool(F)) + MLP(MaxPool(F))) \quad (3)$$

$$M_S(F) = \sigma(f^{7 \times 7}([AvgPool(F); MaxPool(F)])) \quad (4)$$

In Eq. (3), σ represents the sigmoid activation function, MLP denotes a multi-layer perceptron, $AvgPool(\cdot)$ denotes average pooling operation, and $MaxPool(\cdot)$ represents max-pooling operation. In Eq. (4), $f^{7 \times 7}$ represents a convolution operation with the filter size of 7×7 .

Therefore, a lightweight attention module, CBAM, is introduced into C2f, and the C2f-CBAM module is constructed based on this attention mechanism. The C2f module is responsible for fusing high-level semantic features with low-level detail features to obtain richer semantic information. The C2f-CBAM module, after incorporating the attention mechanism, can better achieve fine-grained feature selection and fusion, providing the model with more robust feature representation capabilities, thereby achieving higher object detection accuracy and performance. **Fig. 5** provides an illustration of the structure of the C2f-CBAM module. The CBAM is added after Bottleneck's Add operation and the second Conv of C2f, respectively, which can be considered as cascading multiple CBAM modules, with each module following one or more convolutional layers. As a result, each CBAM module can provide optimized feature maps for subsequent convolutional layers. By cascading multiple CBAM modules, it is possible to progressively enhance the network's focus on important features while diminishing those unrelated to the task.

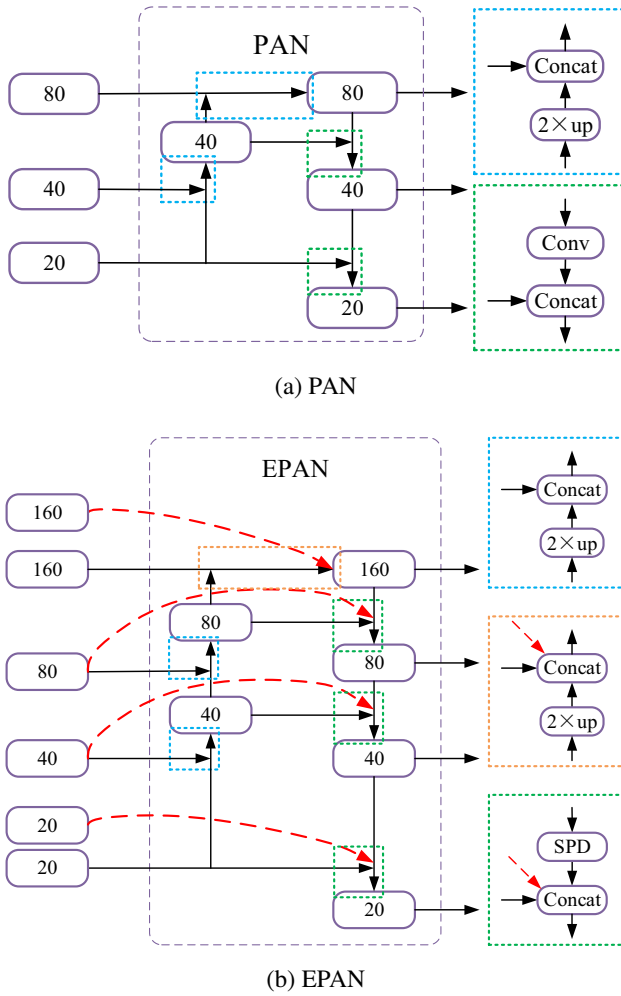


Fig. 6. Structure diagram of PAN and EPAN.

3.4. Proposed Enhanced Path Aggregation Network

YOLOv8 adopts the PAN structure as the neck of the network, as shown in Fig. 6(a). Through upsampling and downsampling operations, it achieves multi-scale feature fusion. However, most of the information of small objects may be lost after being transmitted through multiple layers of the network. Therefore, an optimized design of the network structure is proposed, namely the EPAN, as illustrated in Fig. 6(b). This structure fully integrates low-level features, thereby reducing missed detections and false positives caused by the loss of low-level detail information.

For small objects, their feature information often gets lost in the top feature maps during the transmission process. Therefore, EPAN not only includes a feature fusion component for the newly added small object detection branch but also introduces an additional skip connection between the input and output stages. Compared to the original PAN, the added feature fusion component for the small object detection branch complements and enhances the expressive ability for the semantic features and location information of small objects. The skip connection allows for more comprehensive information transfer between features at different levels, enabling the model to more fully

consider information from different scales when fusing features of various sizes. This improves the model's flexibility and the accuracy of object detection.

The implementation of EPAN is illustrated in Fig. 6(b), where the numbers within the solid-lined boxes represent the size of the output feature maps; for example, the box labeled "80" indicates that the layer outputs a feature map with a size of 80×80 . The arrows indicate the direction of feature map transmission, and the specific details of the fusion method for feature maps of different sizes are shown within the dashed-lined boxes on the right side of the figure. Within the blue dashed-lined box, both the original PAN used in the model and the improved EPAN follow the same approach: the feature map of a smaller size is upsampled by a factor of two to increase its size, thereby enabling concatenation with the feature map from another branch along the channel dimension. Within the orange dashed-lined box, we find the feature fusion component for the newly added small object detection branch, where the feature map of a smaller size is similarly upsampled by a factor of two to increase its size and is concatenated with two feature maps of the same size output from the backbone along the channel dimension. Within the green dashed-lined box, the feature map of a larger size is downsampled by a factor of two to reduce its size, thereby enabling concatenation with the feature map from another branch along the channel dimension. The difference lies in that PAN employs convolution for downsampling, whereas EPAN uses SPD for downsampling. Additionally, compared to PAN, which fuses two feature maps, EPAN further concatenates the feature map extracted from the backbone with these two feature maps along the channel dimension through skip connections, achieving reconstruction of high-dimensional feature maps. This approach can transmit the feature information from the backbone network to the deeper layers of the network, further enhancing the expressive power of the features and enabling more effective target localization.

4. Experimental Results and Analysis

All experiments conducted in this paper employ the PyTorch framework and are run on a server equipped with an Intel Core i9-13900K CPU and an Nvidia GeForce RTX 4080 GPU. In the course of the experiment, none of the networks employed pre-trained weights at the initial training stage; instead, the model weights were randomly initialized. The training configurations of each algorithm use the recommended settings, with the input image size set to 640×640 , a batch size of 4, and a total of 300 training epochs for the algorithm proposed in this paper.

4.1. Dataset

The research presented in this paper employs the publicly available VisDrone2019 dataset for experimental purposes. The images in this dataset were captured under various weather conditions, traffic scenarios, and lighting environments in 14 cities within China. The dataset comprises 6471 images for training, 548 images for validation,

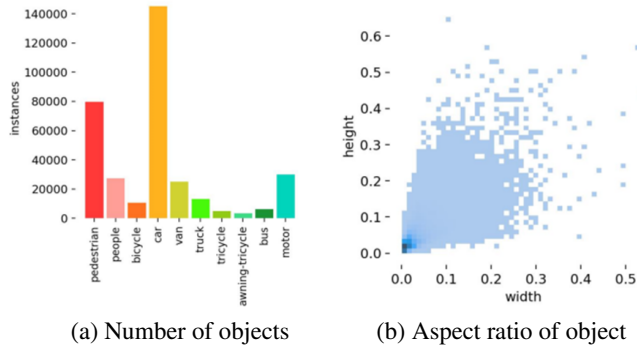


Fig. 7. VisDrone2019 dataset.

and 1610 images for testing. Additionally, 10 types of objects are labeled, as illustrated in Fig. 7(a). The horizontal axis represents the object categories, namely pedestrian, people, bicycle, car, van, truck, tricycle, awning-tricycle, bus, and motor, while the vertical axis indicates the number of samples in each category. Fig. 7(b) illustrates the height-to-width ratio of the objects in the dataset images relative to the entire image. Furthermore, as shown by the color intensity information in the figure, it is evident that the majority of target objects occupy less than one-tenth of the height and width dimensions in the images. This indicates the presence of a substantial number of small objects in the VisDrone2019 dataset, which is well-aligned with the research question explored in this study. Consequently, this study employs the VisDrone2019-DET-train subset of the dataset for model training, utilizes the VisDrone2019-DET-val subset for model validation, and ultimately evaluates the model's performance on the VisDrone2019-DET-test-dev subset.

4.2. Evaluation Metrics

To comprehensively evaluate model performance, we use commonly used metrics such as precision (P), recall (R), mean average precision (mAP), model parameters (Params), floating point operations (FLOPs), and frames per second (fps) for inference speed as evaluation metrics.

Precision represents the proportion of samples that the model correctly predicts as positive among all those predicted as positive, serving as an indicator of the model's prediction accuracy. Recall, on the other hand, quantifies the proportion of actual positive samples that are correctly identified by the model, reflecting its ability to recognize various target categories. The mathematical formulations of precision and recall are presented in Eqs. (5) and (6), respectively.

$$P = \frac{TP}{TP + FP} \quad (5)$$

$$R = \frac{TP}{TP + FN} \quad (6)$$

Here, TP represents true positive, where both the model's prediction and the ground truth are positive; FP denotes false positive, indicating that the model predicts a positive result but the ground truth is negative;

Table 1. Comparison of performance on the VisDrone2019-DET-test-dev subset between Model A (integrated with a P2-scale detection layer) and the original model.

Model	P [%]	R [%]	$mAP50$ [%]	$mAP50-95$ [%]
YOLOv8n	39.1	29.4	27.0	15.1
A	39.7	31.9	29.2	16.5

FN signifies false negative, which occurs when the model predicts a negative result but the ground truth is positive.

The mAP value provides a holistic evaluation metric to assess the model's performance across all categories, as defined in Eq. (7). Specifically, $mAP50$ denotes the average precision across all categories when the intersection over union (IoU) threshold is fixed at 0.5. Additionally, $mAP50-95$ represents the average precision averaged over all categories, with the IoU threshold varying from 0.5 to 0.95.

$$mAP = \frac{1}{n} \sum_{i=1}^n AP(i) \quad (7)$$

$$AP = \int_0^1 P(R) dR \quad (8)$$

In Eq. (7), n signifies the total number of categories, while $AP(i)$ denotes the AP for the i -th category. Eq. (8) defines AP , which is the average precision of a particular category, obtained by calculating the area under the precision-recall ($P(R)$) curve for that category.

Params refers to the total number of learnable weights during the model training process, and this metric is often used to measure the size of the model. FLOPs, on the other hand, represent the computational load of the model and serve as an important criterion for assessing its complexity. Additionally, fps reflects the number of images that the model can process within one second.

4.3. Analysis of Experimental Results

4.3.1. P2-Scale Detection Layer

The introduction of the P2-scale detection layer enhances the model's capability to capture intricate details of small targets by leveraging multi-layer processing and extracting feature information across diverse scales. To evaluate the impact of adding this layer, we constructed Model A incorporating the P2-scale detection layer and conducted a comparative analysis with the baseline model, YOLOv8n, using the VisDrone2019-DET-test-dev subset. The experimental results, presented in Table 1, demonstrate that the incorporation of the P2-scale detection layer enhances the model's performance. Specifically, the P and R values increased by 0.6% and 2.5%, respectively, while the $mAP50$ and $mAP50-95$ increased by 2.2% and 1.4%, respectively. This underscores the effectiveness of the P2-scale detection layer in improving detection outcomes.

Table 2. Comparison of performance on the VisDrone2019-DET-test-dev subset between Model B (integrated with the RFAC-SPD module) and the original model.

Model	<i>P</i> [%]	<i>R</i> [%]	<i>mAP50</i> [%]	<i>mAP50-95</i> [%]
YOLOv8n	39.1	29.4	27.0	15.1
B	43.6	34.0	31.9	18.4

Table 3. Comparison of performance on the VisDrone2019-DET-test-dev subset between Model C (integrated with the C2f-CBAM module) and the original model.

Model	<i>P</i> [%]	<i>R</i> [%]	<i>mAP50</i> [%]	<i>mAP50-95</i> [%]
YOLOv8n	39.1	29.4	27.0	15.1
C	39.1	29.6	27.4	15.3

4.3.2. RFAC-SPD Modules

In YOLOv8n, convolutional downsampling progressively reduces the size of the feature map while expanding the receptive field to capture more abstract and global features. However, this process entails the loss of some fine-grained spatial information, resulting in imprecise target localization. To address this issue, the incorporation of the RFAC-SPD module enhances the model's ability to process intricate details and complex patterns within the image. This module preserves essential target information while effectively reducing the spatial dimension of the feature map. To validate the impact of integrating the RFAC-SPD module within the model, Model B was constructed by substituting the convolutional downsampling in YOLOv8n with the RFAC-SPD module. An experimental comparison with the original model was conducted using the VisDrone2019-DET-test-dev subset. The results, presented in **Table 2**, reveal that Model B exhibits significant improvements in both *P*-value and *R*-value, with increases of 4.5% and 4.6%, respectively. Furthermore, the *mAP50* and *mAP50-95* were enhanced by 4.9% and 3.3%, respectively. These findings underscore the efficacy of the RFAC-SPD module in enhancing the accuracy of object detection, particularly in identifying and locating small objects.

4.3.3. C2f-CBAM Module

The C2f module possesses the capability to integrate feature maps of different scales, ultimately improving the precision of object detection. In this study, we propose the C2f-CBAM module, which is formed by seamlessly integrating the CBAM module into the C2f framework, enabling it to pinpoint crucial features within images with greater accuracy. To assess the efficacy of incorporating the C2f-CBAM module into the baseline YOLOv8n model, we devised Model C by substituting all occurrences of the C2f module with the C2f-CBAM module. We then conducted comparative experiments using the VisDrone2019-DET-test-dev subset, benchmarking Model C against the original YOLOv8n. The results shown in **Table 3** indicate slight improvements in

Table 4. Comparison of performance on the VisDrone2019-DET-test-dev subset between Model D (integrated with the EPAN) and the original model.

Model	<i>P</i> [%]	<i>R</i> [%]	<i>mAP50</i> [%]	<i>mAP50-95</i> [%]
YOLOv8n	39.1	29.4	27.0	15.1
D	40.0	28.8	27.4	15.3

Model C's *R*-value, *mAP50*, and *mAP50-95* by 0.2%, 0.4%, and 0.2%, respectively. These findings partially corroborate the potential of the C2f-CBAM module to enhance the detection performance for small objects.

4.3.4. Enhanced Path Aggregation Network

In the context of YOLOv8n, FPN serves as the neck network, responsible for extracting and integrating features across various scales by constructing a feature pyramid. However, it falls short in adequately capturing and preserving the intricate details and rich semantic content of small targets. To address this limitation, the EPAN is introduced, which appends additional skip connection onto the FPN framework, thereby bolstering the semantic richness of low-level features and refining the efficacy of multi-scale feature fusion. To assess the impact of the EPAN, we formulated Model D by incorporating it as the neck network within YOLOv8n and conducted comparative experiments with the baseline YOLOv8n on the VisDrone2019-DET-test-dev subset. The results presented in **Table 4** indicate that Model D experiences a slight decrease in the *R*-value, but sees an improvement of 0.9% in the *P*-value, along with increases of 0.4% and 0.2% in *mAP50* and *mAP50-95*, respectively. These findings to some extent validate the positive influence of the EPAN on enhancing the precision of small object detection.

4.3.5. Ablation Experiments

Ablation experiments were conducted on the VisDrone2019-DET-test-dev subset using both the proposed algorithm and the YOLOv8n model, and the experimental outcomes are presented in **Table 5**.

Columns 2 to 5 in **Table 5** indicate whether a corresponding structure is added (✓) or not (×). From the results in **Table 5**, it can be observed that the detection performance of the improved model on the VisDrone2019-DET-test-dev subset has been enhanced. The experimental results from model 0 represent the original YOLOv8n model. In Group 1, the P2-scale detection layer was incorporated into the YOLOv8n model. Compared to the original YOLOv8n, the *mAP50* and *mAP50-95* improved by 2.2% and 1.4%, respectively. In Group 2, the RFAC-SPD module and SPD were used to replace the downsampling layers in the backbone and neck of the original network, respectively, based on Group 1's configuration. Although this led to an increase in the number of parameters and computation, the *mAP50* and *mAP50-95* further improved by 5.6% and 3.4% compared to Group 1. In Group 3, the C2f module in the network was replaced with

Table 5. Ablation experimental results on the VisDrone2019-DET-test-dev subset.

Model	P2	RFAC-SPD	C2f-CBAM	EPAN	<i>mAP50</i> [%]	<i>mAP50-95</i> [%]	Params [M]	FLOPs [G]	fps
0	×	×	×	×	27.0	15.1	3.01	8.1	333
1	✓	×	×	×	29.2	16.5	2.92	12.2	277
2	✓	✓	×	×	34.8	19.9	3.79	25.3	78
3	✓	✓	✓	×	34.9	20.0	4.04	25.6	77
4	✓	✓	✓	✓	35.3	20.4	4.13	25.9	77

Table 6. Comparative experimental results on the VisDrone2019-DET-test-dev subset.

Model	<i>mAP50</i> [%]	<i>mAP50-95</i> [%]	Params [M]	FLOPs [G]
Faster RCNN	31.5	17.2	41.2	118.8
YOLOv5s	28.8	15.5	7.04	15.8
YOLOX_s	31.2	18.5	8.94	26.65
YOLOv8s	32.2	18.4	11.1	28.5
YOLOv10s	32	18.5	8.04	24.5
YOLOv10m	34.5	20.4	16.4	63.5
YOLOv11s	32.6	19.1	9.42	21.3
DBAB-YOLO [27]	29.8	16.3	3.2	10.8
YOLO-DroneMS [30]	31.2	–	3.14	–
Drone-YOLO [34]	33.7	19.4	5.35	–
VAMYOLOX-S [25]	34.2	20.17	9.69	60.35
YOLOv8-NDTiny [26]	34.4	20.2	2.3	11.7
YOLO-ERF-S [23]	35.2	18.8	3.7	12.5
YOLOv8-FE	35.3	20.4	4.13	25.9

the C2f-CBAM module on top of Group 2's improvements, resulting in an additional boost of 0.1% both in *mAP50* and *mAP50-95*. Finally, in Group 4, the EPAN was further introduced based on Group 3, leading to further improvements in *mAP50* and *mAP50-95*. As evident from the experimental data in **Table 5**, the proposed modifications led to an overall increase in *mAP50* and *mAP50-95*, ultimately reaching 35.3% and 20.4%, respectively. Despite the increase in Params and FLOPs, the final inference speed reaches 77 frames per second, satisfying the requirement for real-time performance (fps $\geq$ 60). This demonstrates the effectiveness of the improved model.

4.3.6. Comparative Experiments

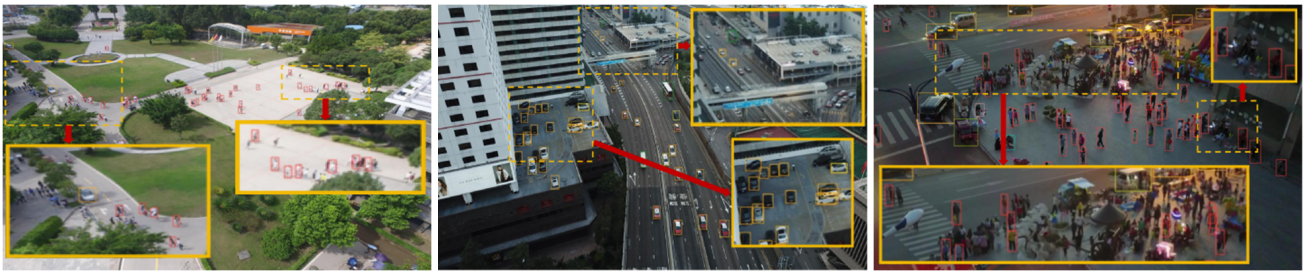
To provide a comprehensive comparison of YOLOv8-FE with other algorithms, this paper contrasts the model's performance on the VisDrone2019-DET-test-dev subset with both current mainstream object detection algorithms and some algorithms specifically proposed for small object detection. The comparative results are presented in **Table 6**.

The results presented in **Table 6** clearly illustrate that, in comparison with other algorithms, the YOLOv8-FE proposed in this paper achieves higher values in both *mAP50* and *mAP50-95*, which compellingly validates its superior detection performance. Although in terms of Params and FLOPs, the algorithm in this paper does not reach the optimal level, it still maintains a relatively outstanding performance. When all evaluation metrics are taken into com-

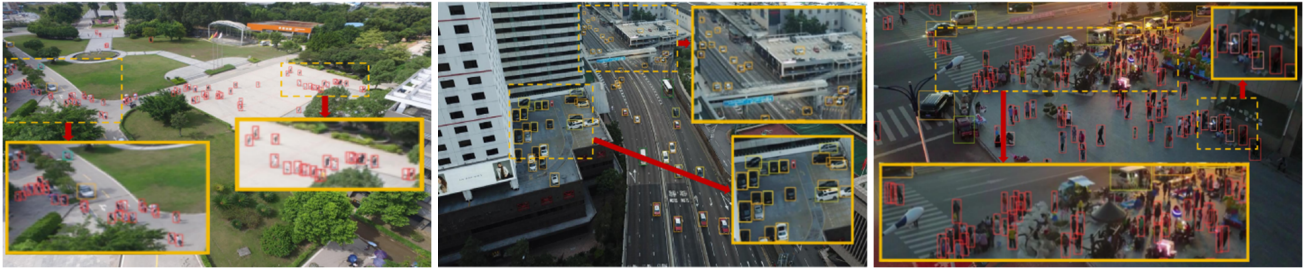
prehensive consideration, the algorithm proposed in this paper showcases distinct superiority.

4.4. Analysis of Visualization Results

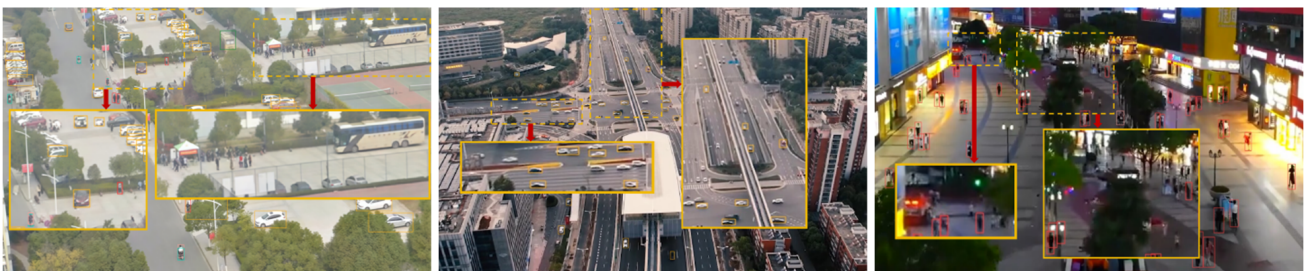
To validate the detection performance of YOLOv8-FE, images under different lighting conditions and scenarios were selected from the VisDrone2019 dataset for testing. For a more visually striking comparison, we applied partial enlargement to the key parts of the images, as shown in **Fig. 8**. This figure presents the detection outcomes of the YOLOv8n model on the first row and those of the YOLOv8-FE model on the second row. Upon inspection of **Fig. 8**, it becomes evident that the refined model proposed in this paper demonstrates enhanced detection capabilities for small object detection tasks, accompanied by a notable decrease in missed detections. Furthermore, to verify the detection performance of YOLOv8-FE in real-world scenarios, images under different lighting conditions and scenarios were captured from publicly available aerial videos for testing, as depicted in **Fig. 9**. This figure displays the detection outcomes of the YOLOv8n model on the first row and those of the YOLOv8-FE model on the second row. In real-world scenarios, it is apparent from **Fig. 9** that the YOLOv8-FE model exhibits a pronounced advantage in detecting small objects, achieving a significantly lower rate of missed detections compared to the YOLOv8n model. Consequently, the enhancements introduced in the YOLOv8-FE model effectively elevate the de-



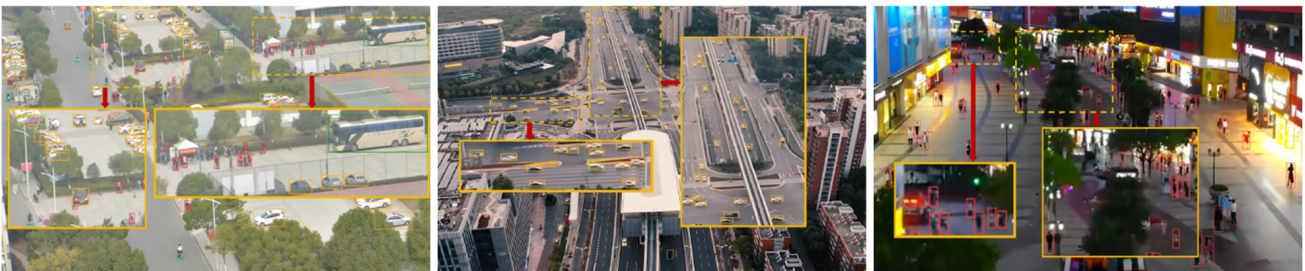
(a) The effect of visualization corresponding to the YOLOv8n



(b) The effect of visualization corresponding to the YOLOv8-FE

Fig. 8. A comparative effect of visualization on the VisDrone2019 dataset before and after improvement.

(a) The effect of visualization corresponding to the YOLOv8n



(b) The effect of visualization corresponding to the YOLOv8-FE

Fig. 9. A comparative effect of visualization in real-world scenarios before and after improvement.

tection performance of the YOLOv8n model for small objects.

4.5. Generalization Experiments

To comprehensively assess the performance of the YOLOv8-FE algorithm across various object detection tasks, we have chosen the DOTA dataset, specifically designed for remote sensing image object detection, and the Caltech Pedestrian dataset, a benchmark for pedestrian detection and tracking, for experiments. The DOTA dataset,

featuring a large number of images with 15 common categories of objects exhibiting diverse scales, orientations, and shapes, presents significant challenges to object detection algorithms. Meanwhile, the Caltech Pedestrian dataset, comprising numerous images captured from diverse scenes and viewpoints, faces multiple hurdles, including occlusion, lighting variations, and intricate backgrounds. The models were all trained on the training set and evaluated on the validation set. The experimental results are summarized in **Table 7**, where “✓” denotes the inclusion of the corresponding structure, while “x” indi-

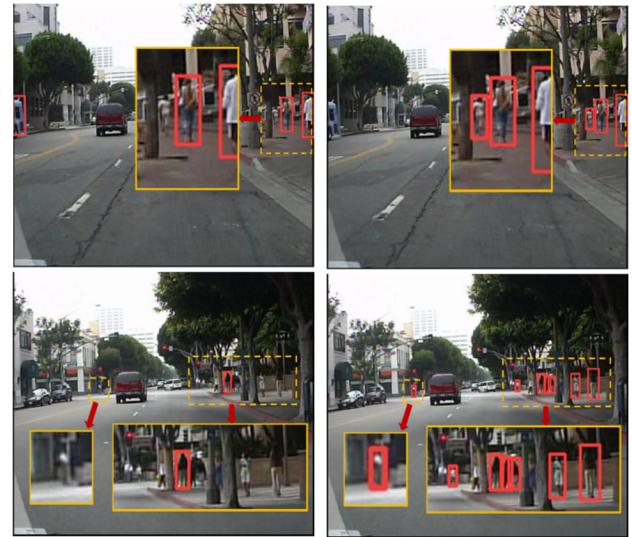
Table 7. Generalization experimental results on the validation sets of the DOTA dataset and Caltech Pedestrian dataset.

Model	P2	RFAC-SPD	C2f-CBAM	EPAN	DOTA		Caltech Pedestrian	
					<i>mAP50</i> [%]	<i>mAP50-95</i> [%]	<i>mAP50</i> [%]	<i>mAP50-95</i> [%]
0	×	×	×	×	59.4	37.0	64.2	40.0
1	✓	×	×	×	58.9	37.2	65.7	42.1
2	✓	✓	×	×	62.3	39.6	70.0	45.5
3	✓	✓	✓	×	62.1	39.3	69.9	45.7
4	✓	✓	✓	✓	62.1	39.7	71.0	46.0



(a) YOLOv8n

(b) YOLOv8-FE

Fig. 10. A comparative effect of detection performance on the DOTA dataset before and after improvement. (a) The left side corresponds to the YOLOv8n. (b) The right side corresponds to the YOLOv8-FE.

(a) YOLOv8n

(b) YOLOv8-FE

Fig. 11. A comparative effect of detection performance on the Caltech Pedestrian dataset before and after improvement. (a) The left side corresponds to the YOLOv8n. (b) The right side corresponds to the YOLOv8-FE.

cates its exclusion.

As is evident from **Table 7**, the diverse improved modules proposed in this paper demonstrate favorable generalization capabilities across different datasets, albeit with room for further improvement. In general, on the DOTA dataset, the improved YOLOv8-FE algorithm achieves an enhancement of 2.7% in both *mAP50* and *mAP50-95* compared to the original YOLOv8n model. Regarding the Caltech Pedestrian detection dataset, the improved YOLOv8-FE algorithm demonstrates an increase of 6.8% in *mAP50* and 6.0% in *mAP50-95* compared to the baseline, respectively. This demonstrates that the YOLOv8-FE algorithm possesses good universality and adaptability, making it suitable for datasets with large scale variations and complex backgrounds.

Figures 10 and 11 illustrate a comparative analysis of the detection results obtained by the YOLOv8n model and the optimized YOLOv8-FE model on the DOTA dataset and the Caltech Pedestrian detection dataset, respectively. Specifically, the left-hand side of both figures displays the detection outcomes of the YOLOv8n model, whereas the right-hand side corresponds to the detection results of the enhanced YOLOv8-FE model. Upon comparison, a no-

table superiority in detection performance is evident in the improved YOLOv8-FE model, as opposed to the original YOLOv8n model. This observation robustly validates the efficacy of the optimized model and underscores its impressive generalization capabilities.

5. Conclusion

To address the common challenges of low detection accuracy and high miss rates in small object detection, this paper introduces an enhanced algorithm, YOLOv8-FE, which builds upon the YOLOv8 framework. Initially, we augment the network with a P2-scale detection layer to improve its capability in detecting small objects. Subsequently, an RFAC-SPD layer is incorporated to mitigate information loss during the downsampling process. Furthermore, we design a C2f-CBAM module to replace the existing C2f module, thereby enhancing the model's ability to extract pertinent target features. Lastly, the EPAN structure is proposed to facilitate the fusion of diverse input features within the network. Experimental results on the VisDrone2019 dataset reveal that our proposed algorithm

attains mAP50 and mAP50-95 scores of 35.3% and 20.4% respectively, showcasing superior detection accuracy compared to other models. Additionally, the algorithm in this paper achieves an inference speed of 77 frames per second, satisfying the requirements for real-time performance.

Although experimental results demonstrate that the algorithm proposed in this paper has achieved significant improvements in enhancing small target detection capabilities, the enhancement in detection capability is accompanied by an increase in network parameters and computational load, posing challenges to the real-time performance and deployment efficiency of the algorithm, especially in resource-constrained application scenarios. Consequently, striking a reasonable balance between algorithm complexity and computational efficiency while pursuing higher detection accuracy has emerged as a crucial issue in current research. Looking ahead, our research objectives will focus on further boosting detection accuracy and will also be dedicated to optimizing the computational speed of the model, with the aim of developing a more lightweight model that strikes an optimal balance between speed and accuracy.

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Name:
Mingxing Fang

Affiliation:
School of Physics and Electronic Information,
Anhui Normal University

Address:

No.189 Jiuhua South Road, Yijiang District, Wuhu, Anhui 241002, China

Brief Biographical History:

1995-2000 Assistant, Anhui Normal University
2000-2007 Lecturer, Anhui Normal University
2007-2017 Associate Professor, Anhui Normal University
2009 Received Ph.D. degree from School of Information Science and Engineering, Central South University
2017- Professor, Anhui Normal University

Main Works:

- "Improving disturbance-rejection performance based on an equivalent-input-disturbance approach," IEEE Trans. on Industrial Electronics, Vol.55, No.1, pp. 380-389, 2008.

Membership in Academic Societies:

- The China Automation Association



Name:
Hongyu Cheng

Affiliation:
School of Physics and Electronic Information,
Anhui Normal University

Address:

No.189 Jiuhua South Road, Yijiang District, Wuhu, Anhui 241002, China

Brief Biographical History:

2020-2022 Bachelor's Student, Huainan Normal University
2022- Masters Student, Anhui Normal University



Name:
Xinke Liu

Affiliation:
School of Physics and Electronic Information,
Anhui Normal University

Address:

No.189 Jiuhua South Road, Yijiang District, Wuhu, Anhui 241002, China

Brief Biographical History:

2018-2022 Bachelor's Student, Wanjiang College, Anhui Normal University
2022- Master's Student, Anhui Normal University



Name:
Xinyu Rui

Affiliation:
School of Physics and Electronic Information,
Anhui Normal University

Address:

No.189 Jiuhua South Road, Yijiang District, Wuhu, Anhui 241002, China

Brief Biographical History:

2016-2020 Bachelor's Student, Anhui Normal University
2022- Master's Student, Anhui Normal University



Name:
Jinhua She

Affiliation:
School of Engineering, Tokyo University of Technology

Address:

1404-1 Katakuramachi, Hachioji, Tokyo 192-0982, Japan

Brief Biographical History:

1993 Received Ph.D. degree in Engineering from Tokyo Institute of Technology

1993-2001 Lecturer, Tokyo University of Technology

2001-2010 Associate Professor, Tokyo University of Technology

2010- Professor, Tokyo University of Technology

Main Works:

- “Disturbance rejection and control system design using {1}-inverse-based equivalent-input-disturbance approach,” IEEE Trans. on Industrial Electronics, Vol.70, No.2, pp. 1666-1675, 2023.
- “Assistive robots,” M. Wu, W. Pedrycz, and L. Chen (Eds.), “Developments in Advanced Control and Intelligent Automation for Complex Systems,” pp. 361-392, Springer, 2021.

Membership in Academic Societies:

- The Society of Instrument and Control Engineers (SICE)
- The Institute of Electrical Engineers of Japan (IEEJ)
- The Japan Society of Mechanical Engineers (JSME)
- Chinese Association of Automation (CAA)
- Asian Control Association (ACA)
- Institute of Electrical and Electronics Engineers (IEEE)



Name:
Haoran Tan

Affiliation:
College of Electrical and Information Engineering / National Engineering Research Center of Robot Visual Perception and Control Technology, Hunan University

Address:

No.2 Lushan South Road, Yuelu District, Changsha, Hunan 410082, China

Brief Biographical History:

2021- Associate Professor, College of Electrical and Information Engineering, Hunan University



Name:
Youwu Du

Affiliation:
School of Electrical and Information Engineering, Jiangsu University of Technology

Address:

No.1801 Zhongwu Road, Changzhou, Jiangsu 213001, China

Brief Biographical History:

2010 Received M.E. degree in Control Science and Engineering from Central South University

2020 Received Ph.D. degree from China University of Geosciences